

MA341

APPLIED ODE'S

Homework 3.5

1. Use Euler's Method for Linear Systems of ODE to find **real valued** set of fundamental solutions

$$(a) \frac{d}{dt}X = \begin{pmatrix} 1 & -2 \\ 3 & -4 \end{pmatrix} X \quad (b) \frac{d}{dt}X = \begin{pmatrix} 4 & -3 \\ 8 & -6 \end{pmatrix} X \quad (c) \frac{d}{dt}X = \begin{pmatrix} 4 & 3 \\ -6 & -2 \end{pmatrix} X$$

Solution:

- (a) Eigenvalues and their associated eigenvectors:

$$\lambda_1 = -2, v_1 = \begin{pmatrix} 2 \\ 3 \end{pmatrix} \quad \lambda_2 = -1, v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Hence the general solution is

$$x(t) = c_1 x_1(t) + c_2 x_2(t) = c_1 \begin{pmatrix} 2 \\ 3 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-t}$$

- (b) Eigenvalues and their associated eigenvectors:

$$\lambda_1 = -2, v_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad \lambda_2 = 0, v_2 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

Hence the general solution is

$$x(t) = c_1 x_1(t) + c_2 x_2(t) = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

- (c) Eigenvalues and their associated eigenvectors:

$$\lambda_{1,2} = 1 \pm 3i, v_{1,2} = \begin{pmatrix} -1 \mp i \\ 2 \end{pmatrix}$$

Denote

$$\begin{aligned} \tilde{x}(t) &= \begin{pmatrix} -1-i \\ 2 \end{pmatrix} e^{(1+3i)t} = \begin{pmatrix} (-1-i)(\cos 3t + i \sin 3t) \\ 2(\cos 3t + i \sin 3t)e^t \end{pmatrix} \\ &= \begin{pmatrix} -\cos 3t + \sin 3t \\ 2 \cos 3t \end{pmatrix} e^t + i \begin{pmatrix} -\cos 3t - \sin 3t \\ 2 \sin 3t \end{pmatrix} e^t \end{aligned}$$

Hence the general solution is

$$\begin{aligned} x(t) &= c_1 x_1(t) + c_2 x_2(t) \\ &= c_1 \operatorname{Re}(\tilde{x}(t)) + c_2 \operatorname{Im}(\tilde{x}(t)) \\ &= c_1 \begin{pmatrix} -\cos 3t + \sin 3t \\ 2 \cos 3t \end{pmatrix} e^t + c_2 \begin{pmatrix} -\cos 3t - \sin 3t \\ 2 \sin 3t \end{pmatrix} e^t \end{aligned}$$

2. Use Euler's Method for Linear Systems of ODE to solve the following initial value problems.

(a) $\frac{d}{dt}X = \begin{pmatrix} 1 & -1 \\ 5 & -3 \end{pmatrix} X, \quad X(0) = \begin{pmatrix} -1 \\ -5 \end{pmatrix}$

Solution: General solution

$$x(t) = c_1 \begin{pmatrix} 2 \cos(t) - \sin(t) \\ 5 \cos(t) \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} 2 \sin(t) + \cos(t) \\ 5 \sin(t) \end{pmatrix} e^{-t}$$

Use ICs

$$x(0) = c_1 \begin{pmatrix} 2 \\ 5 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 5 & 0 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} -1 \\ -5 \end{pmatrix}$$

to solve for $c_1 = -1$ and $c_2 = 1$. Thus,

$$x(t) = - \begin{pmatrix} 2 \cos(t) - \sin(t) \\ 5 \cos(t) \end{pmatrix} e^{-t} + \begin{pmatrix} 2 \sin(t) + \cos(t) \\ 5 \sin(t) \end{pmatrix} e^{-t} = \begin{pmatrix} 3 \sin(t) - \cos(t) \\ 5 \sin(t) - 5 \cos(t) \end{pmatrix} e^{-t}$$

(b) $\frac{d}{dt}X = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 1 \end{pmatrix} X, \quad X(0) = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}$

Solution:

$$x(k) = c_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} e^{-t} + c_3 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} e^t.$$

Use ICs

$$x(k) = c_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} + c_3 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}.$$

To solve for $c_1 = 1$, $c_2 = -1$ and $c_3 = 0$. Thus,

$$x(k) = c_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} e^{3t} - c_2 \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} e^{-t}.$$

$$(c) \quad \frac{d}{dt}X = \begin{pmatrix} 4 & 0 & -3 \\ -3 & 1 & 3 \\ 6 & 0 & -5 \end{pmatrix} X, \quad X(0) = \begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix}$$

Solution: General solution

$$x(k) = c_1 \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} e^t + c_3 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} e^t.$$

Use ICs

$$x(0) = c_1 \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + c_3 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix}$$

to solve for $c_1 = 1$, $c_2 = 1$ and $c_3 = 1$. Thus,

$$x(k) = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} e^{-2t} + \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} e^t + \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} e^t.$$

3. Consider $y'''(t) - y''(t) - y'(t) + y(t) = 0$

(a) Denote new variables $x_1(t) := y(t)$, $x_2(t) := y'(t)$, $x_3(t) := y''(t)$ and solve the following system

$$X'(t) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 1 \end{pmatrix} X(t), \quad X(t) := \begin{pmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{pmatrix} = \begin{pmatrix} y(t) \\ y'(t) \\ y''(t) \end{pmatrix}$$

Solution:

$$X(k) = c_1 \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} e^t + c_3 \left(\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} t + \begin{pmatrix} -2 \\ -1 \\ 0 \end{pmatrix} \right) e^t$$

(b) Use your solution to the system to find the solution to the original equation (verify!).

Solution:

$$\begin{aligned} y(t) = x_1(t) &= c_1 e^{-t} + c_2 e^t + c_3(t-2)e^t \\ &= c_1 e^{-t} + (c_2 + c_3(t-2))e^t \\ &= c_1 e^{-t} + \underbrace{(c_2 - 2c_3)}_{=c_2} + c_3 t e^t \end{aligned}$$

4. Consider the following initial value problem:

$$t\mathbf{x}'(t) = \begin{bmatrix} 3 & -2 \\ 2 & -2 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} -2t \\ t^4 - 1 \end{bmatrix}, \mathbf{x}(1) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

- (a) Show that $\mathbf{x}_1(t) = \begin{bmatrix} t^{-1} \\ 2t^{-1} \end{bmatrix}$ and $\mathbf{x}_2(t) = \begin{bmatrix} 2t^2 \\ t^2 \end{bmatrix}$ are the fundamental solutions to the associated homogeneous system.

Solution: To form the fundamental set $x_1(t)$ and $x_2(t)$ needs to satisfy the equation, and be linearly independent. They are linearly independent since they aren't collinear ($x_1(t) \neq \alpha x_2(t)$). Finally, they satisfy the equation:

$$tx_1'(t) = t \begin{pmatrix} \frac{1}{t} \\ \frac{2}{t} \end{pmatrix}' = t \begin{pmatrix} -\frac{1}{t^2} \\ -\frac{2}{t^2} \end{pmatrix} = \begin{pmatrix} -\frac{1}{t} \\ -\frac{2}{t} \end{pmatrix} = \begin{pmatrix} 3 & -2 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} \frac{1}{t} \\ \frac{2}{t} \end{pmatrix}$$

$$tx_2'(t) = t \begin{pmatrix} 2t^2 \\ t^2 \end{pmatrix}' = t \begin{pmatrix} 4t \\ 2t \end{pmatrix} = \begin{pmatrix} 4t^2 \\ 2t^2 \end{pmatrix} = \begin{pmatrix} 3 & -2 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} 2t^2 \\ t^2 \end{pmatrix}$$

Note: A student offered an interesting, but unfortunately wrong solution, we all can learn from it. We learned a theorem that says that given that $x_1(t)$, $x_2(t)$ are 2 linearly independent solutions to ODE, then Wronskian of $x_1(t)$, $x_2(t)$ satisfies $W'(t) = \text{tr}(A)W(t)$. If so, why verifying this property for the offered solution, doesn't prove they are actually solves the problem? Simply because the converse theorem is not true.

Here is counter example, consider two functions $\begin{pmatrix} 2 \\ 3 \end{pmatrix} e^{11t}$ and $\begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{-4t}$. Their Wronskian, $W = 5e^{7t}$, is definitely satisfies $W' = 7W$. However, they are not the solutions of $x' = \begin{pmatrix} 2 & 6 \\ 3 & 5 \end{pmatrix} x$.

- (b) Use variation of parameters for systems to find the solution to the problem.

Solution: Denote $\Psi(t) = \begin{pmatrix} \frac{1}{t} & 2t^2 \\ \frac{2}{t} & t^2 \end{pmatrix}$, $g(t) = \frac{1}{t} \begin{pmatrix} -2t \\ t^4 - 1 \end{pmatrix} = \begin{pmatrix} -2 \\ \frac{t^4-1}{t} \end{pmatrix}$

$$x_p(t) = \Psi(t) \int \Psi^{-1}(t)g(t)dt = \Psi(t) \int \begin{pmatrix} -\frac{t}{3} & \frac{2t}{3} \\ \frac{2}{3t^2} & -\frac{1}{3t^2} \end{pmatrix} \begin{pmatrix} -2 \\ \frac{t^4-1}{t} \end{pmatrix} dt$$

$$= \Psi(t) \int \begin{pmatrix} \frac{2}{3}(t^4 - 1) + \frac{2t}{3} \\ -\frac{t^4-1}{3t^3} - \frac{4}{3t^2} \end{pmatrix} dt = \Psi(t) \begin{pmatrix} \frac{2}{3} \left(\frac{t^5}{5} + \frac{t^2}{2} - t \right) \\ \frac{1}{3} \left(-\frac{t^2}{2} - \frac{1}{2t^2} + \frac{4}{t} \right) \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{t^4}{5} + 3t - 1 \\ \frac{1}{10}(t^4 + 20t - 15) \end{pmatrix}$$

Thus, the general solution is given by

$$x(t) = c_1 \begin{pmatrix} \frac{1}{t} \\ \frac{2}{t} \end{pmatrix} + c_2 \begin{pmatrix} 2t^2 \\ t^2 \end{pmatrix} + \begin{pmatrix} -\frac{t^4}{5} + 3t - 1 \\ \frac{1}{10}(t^4 + 20t - 15) \end{pmatrix}$$

Use ICs:

$$x(1) = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + c_2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} -\frac{1}{5} + 3 - 1 \\ \frac{1}{10}(1 + 20 - 15) \end{pmatrix} = \begin{pmatrix} c_1 + 2c_2 + \frac{9}{5} \\ 2c_1 + c_2 + \frac{3}{5} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

To get $c_1 = \frac{1}{5}$, $c_2 = -1$.

Thus,

$$x(t) = \frac{1}{5} \begin{pmatrix} \frac{1}{t} \\ \frac{2}{t} \end{pmatrix} - \begin{pmatrix} 2t^2 \\ t^2 \end{pmatrix} + \begin{pmatrix} -\frac{t^4}{5} + 3t - 1 \\ \frac{1}{10}(t^4 + 20t - 15) \end{pmatrix}$$

Substitute it to the original equation to verify it is indeed a solution.

5. Solve the following problems. **Note:** In those problems the geometric multiplicity is less than algebraic multiplicity.

(a) $\frac{d}{dt}X = \begin{pmatrix} 1 & -4 \\ 4 & -7 \end{pmatrix} X \quad X(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}$

Solution: The eigenvalue $\lambda = -3$ has a algebraic multiplicity 2, but geometric multiplicity 1. Thus, the only eigenvector we have is $v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. To find another linearly independent solution solve

$(A + 3I)v_2 = v_1$ for $v_2 = \frac{1}{4} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Hence, the general solution is

$$x(t) = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-3t} + c_2 \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix} t + \frac{1}{4} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right) e^{-3t}$$

Disregard the "IC" it is not really given here.

(b) $\frac{d}{dt}X = \begin{pmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 3 & 6 & 2 \end{pmatrix} X, \quad X(0) = \begin{pmatrix} -1 \\ 2 \\ -30 \end{pmatrix}$

Solution: Eigenvalues and eigenvectors:

$$\lambda_1 = 2, v_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad \lambda_{2,3} = 1, v_2 = \begin{pmatrix} 0 \\ -1 \\ 6 \end{pmatrix}$$

To find another linearly independent solution solve $(A - I)v_3 = v_2$ for $v_3 = \frac{1}{8} \begin{pmatrix} 2 \\ 7 \\ 0 \end{pmatrix}$. Hence, the general solution is

$$x(t) = c_1 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} 0 \\ -1 \\ 6 \end{pmatrix} e^t + c_3 \left(\begin{pmatrix} 0 \\ -1 \\ 6 \end{pmatrix} t + \frac{1}{8} \begin{pmatrix} 2 \\ 7 \\ 0 \end{pmatrix} \right) e^t$$

Use ICs

$$x(0) = c_1 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ -1 \\ 6 \end{pmatrix} + c_3 \frac{1}{8} \begin{pmatrix} 2 \\ 7 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1/4 \\ 0 & -1 & 7/8 \\ 1 & 6 & 0 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \\ -30 \end{pmatrix}$$

to solve for $\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 3 \\ -\frac{11}{2} \\ -4 \end{pmatrix}$ to get

$$x(t) = 3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} e^{2t} - \frac{11}{2} \begin{pmatrix} 0 \\ -1 \\ 6 \end{pmatrix} e^t - 4 \left(\begin{pmatrix} 0 \\ -1 \\ 6 \end{pmatrix} t + \frac{1}{8} \begin{pmatrix} 2 \\ 7 \\ 0 \end{pmatrix} \right) e^t$$